15/05 namespace

namespace is use to create a declarative region for identifiers,so that we can avoid naming collisions.

using namespace std;

directive keyword is a predefined namespace of C++

std ----> standard

what is the role of using?

"using " will get the visibility of members of std to the program

or

using will add all members of std as global members to the program

What std contains?

std contains

1. predefined objects declarations.
2. predefined member function and functions definitions.

What header file contains?

Header file contains

function declarations

class declarations

"using" keyword will add all members to the program , to add only specific members we can use scope resolution operator

ex :

```
#include<iostream>
using namespace std;

int main()
{   int n1=10,n2=20;
cout<<n1<<" "<<n2<<endl;
swap(n1,n2);
cout<<n1<<" "<<n2<<endl;
}
```

ex:

```
#include<iostream>
int main()
{ int n1=10,n2=20;
std::cout<<n1<<" "
<<n2<<std::endl;

std::swap(n1,n2);
std::cout<<n1<<" "
<<n2<<std::endl;
}
```

syntax to create the namespace

```
namespace tagname
{
      members of namespace

}
```

members of namespace can be variables functions classes struct anything else

generic Programming

Generic Programming is the one of the most powerfull feature of C++.
It is obtained using templates in C++.

templates are of two types
1. function templates — are use to create generic functions
2. class templates — are use to create generic classes

syntax to create templates

```
template<class genericdatatype>
```

ex :

```
template<class any>
```

↳ Here "any" is a generic datatype
it's not a keyword or predefined datatype, it is the name given by us.
we can also x, y, P, any name

template created is applicable only to the immediate class or function defined.